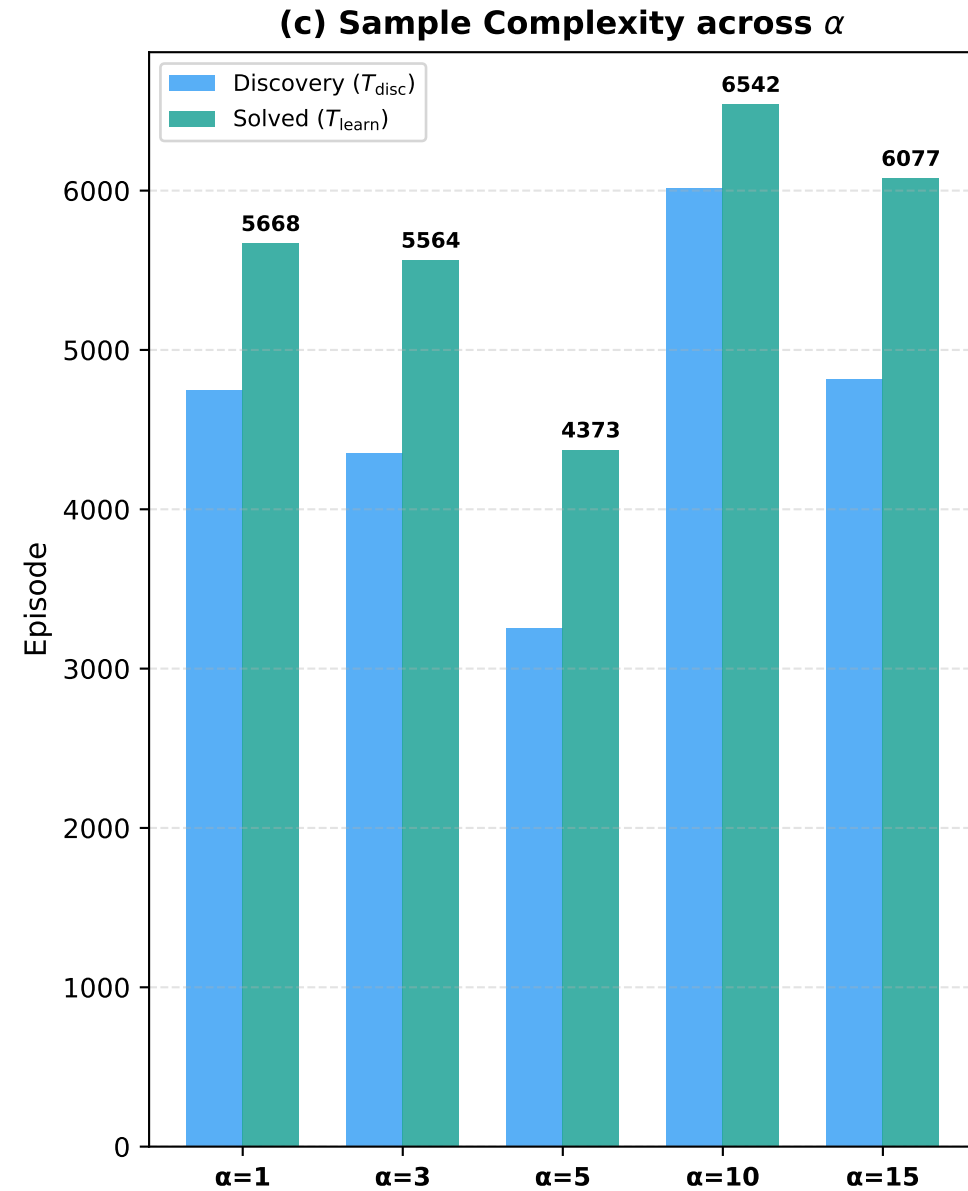
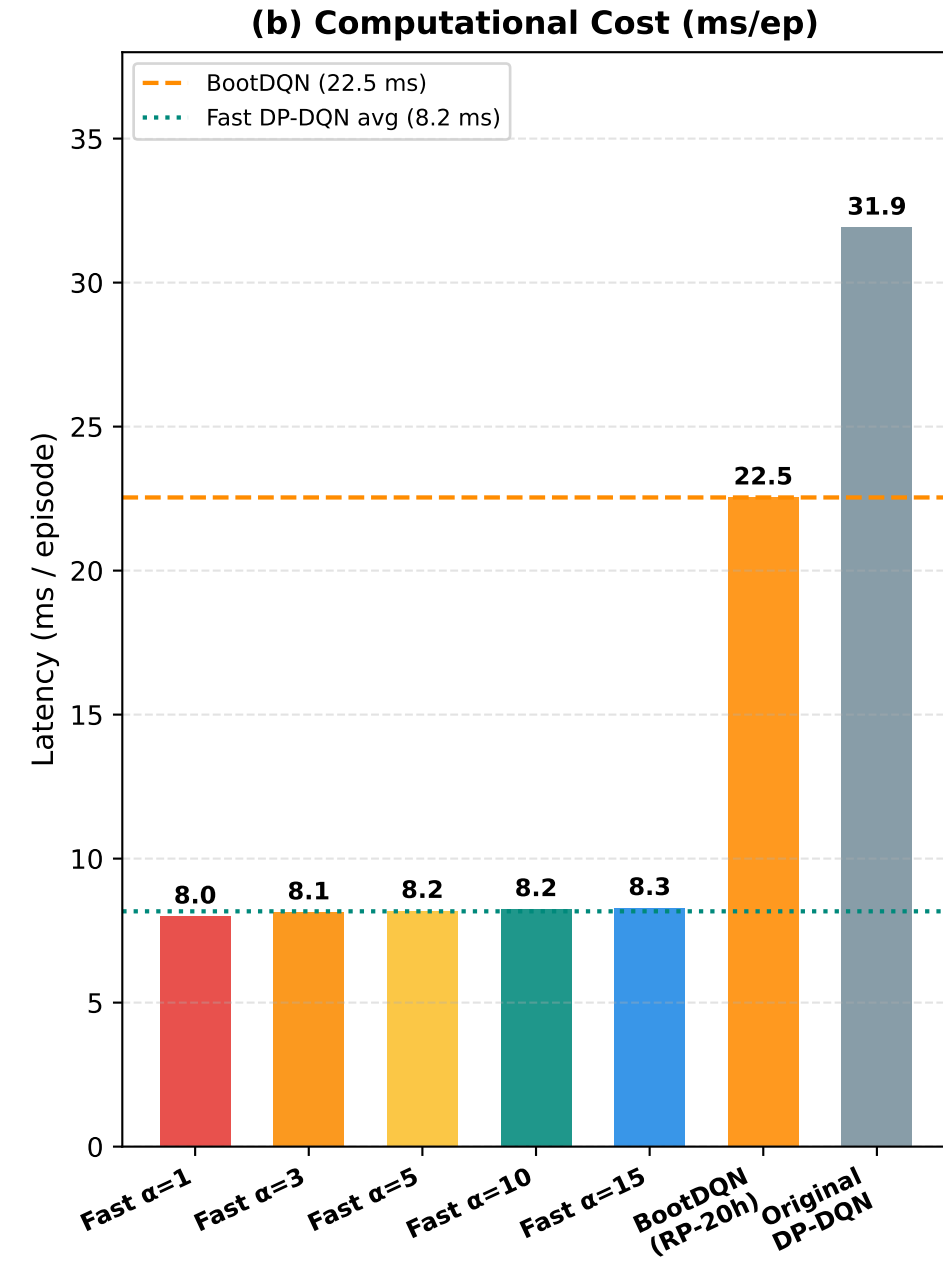
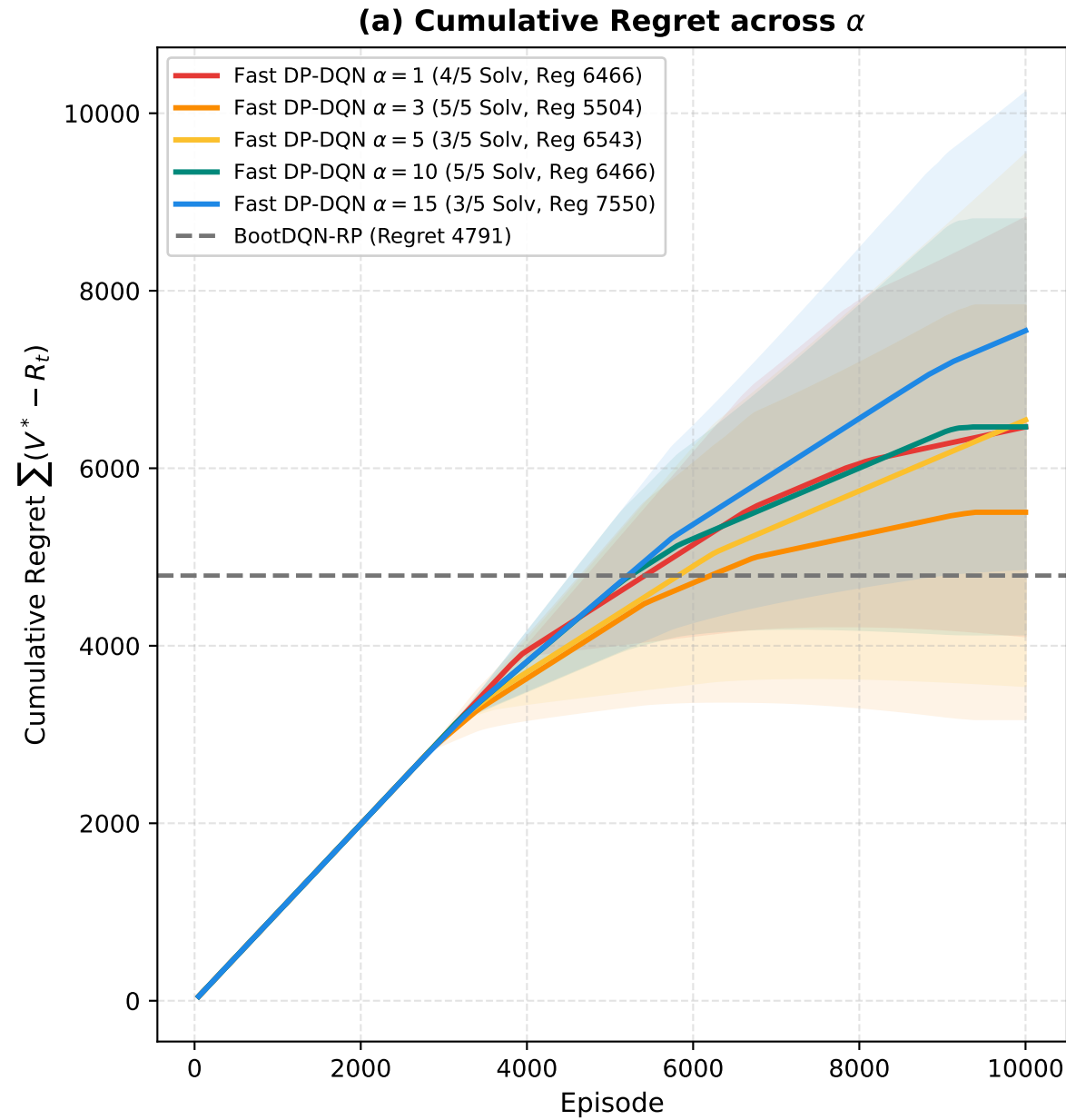


Fast DP-DQN Optimization & Alpha Spectrum ($\alpha \in [1, 15]$) on Deep Sea 20 $\times$ 20



(d) Performance & Speedup Leaderboard

Configuration	Batch (B)	Warmstart	Solved Rate	Mean Solve	Mean Regret	Cost (ms/ep)	Speedup vs BootDQN
Fast DP-DQN ($\alpha=1$)	64	2 steps	4/5 (80%)	5668	6466	8.0 ms	2.8x faster
Fast DP-DQN ($\alpha=3$)	64	2 steps	5/5 (100%)	5564	5504	8.1 ms	2.8x faster
Fast DP-DQN ($\alpha=5$)	64	2 steps	3/5 (60%)	4373	6543	8.2 ms	2.8x faster
Fast DP-DQN ($\alpha=10$)	64	2 steps	5/5 (100%)	6542	6466	8.2 ms	2.7x faster
Fast DP-DQN ($\alpha=15$)	64	2 steps	3/5 (60%)	6077	7550	8.3 ms	2.7x faster
BootDQN + RP (20h)	64	0 steps	5/5 (100%)	5086	4791	22.5 ms	1.0x (Baseline)
Original DP-DQN ($\alpha=10$)	200	5 steps	5/5 (100%)	2995	2909	31.9 ms	0.7x (Slower)